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Phần 1: Các lệnh cơ bản thao tác với Docker

1 docker --version

```
PS C:\Users\Student> docker --version
Docker version 24.0.6, build ed223bc
```

2 docker run hello-world

```
PS C:\Users\Student> docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.
```

3 docker pull nginx

```
PS C:\Users\Student> docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
```

4 docker images

```
PS C:\Users\Student\Phuong> docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest    0cf1d6af5ca7   14 hours ago   161MB
hello-world   latest    e2ac70e7319a   39 hours ago   10.1kB
```

5 docker run -d nginx

```
PS C:\Users\Student\Phuong> docker run -d nginx
876c55bce647f34cffec36968bfd6fb1dfbd04ff746953148373b742494c79b6
```

6 docker ps

```
PS C:\Users\Student\Phuong> docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS     NAMES
876c55bce647   nginx     "/docker-entrypoint.…"   9 seconds ago   Up 5 seconds   80/tcp    elated_wilbur
```

7 docker ps -a

```
PS C:\Users\Student\Phuong> docker ps -a
CONTAINER ID   IMAGE         COMMAND                  CREATED          STATUS                      PORTS     NAMES
876c55bce647   nginx         "/docker-entrypoint.…"   40 seconds ago   Up 36 seconds               80/tcp    elated_wilbu
r
20e613adb7d4   hello-world   "/hello"                 5 minutes ago    Exited (0) 5 minutes ago              crazy_prosku
riakova
19319027ab38   hello-world   "/hello"                 6 minutes ago    Exited (0) 6 minutes ago              nice_robinso
```

8 docker logs <container_id>

```
PS C:\Users\Student\Phuong> docker logs 19319027ab38

Hello from Docker!
This message shows that your installation appears to be working correctly.
```

9 docker exec -it <container_id> /bin/sh

```
PS C:\Users\Student\Phuong> docker exec -it 876c55bce647  /bin/sh
# ls
bin   dev                  docker-entrypoint.sh  home  lib64  mnt  proc  run   srv  tmp  var
boot  docker-entrypoint.d  etc                   lib   media  opt  root  sbin  sys  usr
```

10 docker stop <container_id>

```
PS C:\Users\Student\Phuong> docker stop 19319027ab38
19319027ab38
```

11 docker restart <container_id>

```
PS C:\Users\Student\Phuong> docker restart 876c55bce647
876c55bce647
```

12 docker rm <container_id>

```
PS C:\Users\Student\Phuong> docker rm 876c55bce647 -f
876c55bce647
```

13 docker container prune

```
PS C:\Users\Student\Phuong> docker container prune
WARNING! This will remove all stopped containers.
Are you sure you want to continue? [y/N]
Total reclaimed space: 0B
```

14 docker rmi <image_id>

```
PS C:\Users\Student\Phuong> docker rmi 0cf1d6af5ca7
Untagged: nginx:latest
Untagged: nginx@sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
```

15 docker image prune -a

```
PS C:\Users\Student\Phuong> docker image prune -a
WARNING! This will remove all images without at least one container associated to them.
Are you sure you want to continue? [y/N]
Total reclaimed space: 0B
```

16 docker run -d -p 8080:80 nginx

```
PS C:\Users\Student\Phuong> docker run -d -p 8080:80 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
ec781dee3f47: Pull complete
bb3d0aa29654: Pull complete
510ddf6557d6: Pull complete
cde7a05ae428: Pull complete
587e3d84dbb5: Pull complete
3189680c601f: Pull complete
5e815e07e569: Pull complete
Digest: sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
Status: Downloaded newer image for nginx:latest
eea1c2ec815986d054b810f78c399d3291692e811007dde0d64b45292480d344
```

17 docker inspect <container_id>

```
PS C:\Users\Student\Phuong> docker inspect eea1c2ec815986d054b810f78c399d3291692e811007dde0d64b45292480d344
[
    {
        "Id": "eea1c2ec815986d054b810f78c399d3291692e811007dde0d64b45292480d344",
        "Created": "2026-03-25T12:40:22.5167449Z",
        "Path": "/docker-entrypoint.sh",
        "Args": [
            "nginx",
            "-g",
            "daemon off;"
        ],
```

18 docker run -d -v mydata:/data nginx

```
PS T:\Docker_Phuong> docker run -d -v mydata:/data nginx
341752db04f20b20920f680b06837e1cdc848f9125b8222a6934871976c13df2
```

19 docker volume ls

```
PS T:\Docker_Phuong> docker volume ls
DRIVER    VOLUME NAME
local     mydata
```

20 docker volume prune

```
PS T:\Docker_Phuong> docker volume prune
WARNING! This will remove anonymous local volu
er.
Are you sure you want to continue? [y/N]
Total reclaimed space: 0B
```

21 docker run -d --name my_nginx nginx

```
PS T:\Docker_Phuong> docker run -d --name my_nginx nginx
2e9212ef671b69a732b8b983e6043fdd872bcf447652762108dfcab9385feeb4
```

22 docker stats

```
PS T:\Docker_Phuong> docker stats
```

23 docker network ls

```
PS T:\Docker_Phuong> docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
f1404928be36   bridge    bridge    local
3ca57c3e562c   host      host      local
4ef757842cb0   none      null      local
```

24 docker network create my_network

```
PS T:\Docker_Phuong> docker network create my_network
5fc11187f75a965314c80bff2039b8047f7028deb47f50d0c79d6709df515653
```

25 docker run -d --network my_network --name my_container nginx

```
PS T:\Docker_Phuong> docker run -d --network my_network --name my_container nginx

7d0b76bfd707478cfcfe66608d51d7e4b2a6351d1e3bca47327f514aa2b6d7e9
```

26 docker network connect my_network my_nginx

```
PS T:\Docker_Phuong> docker network connect my_network my_nginx
```

27 docker run -d -e MY_ENV=hello_world nginx

```
PS T:\Docker_Phuong> docker run -d -e MY_ENV=hello_world nginx
7f7e5e174abce697ed83ef6a3f4f3e217e5c6c594c2bdcbb45b66b7bca5d83fa
```

28 docker logs -f my_nginx

```
PS T:\Docker_Phuong> docker logs -f my_nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform
 configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default
.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/d
efault.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d
/default.conf
```

29 FROM nginx

COPY index.html /usr/share/nginx/html/index.html

```
Lab4.docx    Dockerfile 1, U ×    index.html 3, U
Lab4 > Ex30 > Dockerfile > ...
1  FROM nginx
2
3  COPY index.html /usr/share/nginx/html/index.html
```

30 docker build -t my_nginx_image .

```
D:\HK2_2026\KTHT\KTTKPM_HuynhThanhPhuong_22671361_DHKTPM18ATT\Lab4\Ex30 [main ≡ +2 ~0 -0 !]> docker build -t my_nginx_image .
[+] Building 1.0s (7/7) FINISHED                                                                          docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                       0.0s
 => => transferring dockerfile: 99B                                                                                        0.0s
 => [internal] load metadata for docker.io/library/nginx:latest                                                            0.0s
 => [internal] load .dockerignore                                                                                          0.0s
 => => transferring context: 2B                                                                                            0.0s
 => [internal] load build context                                                                                          0.1s
```

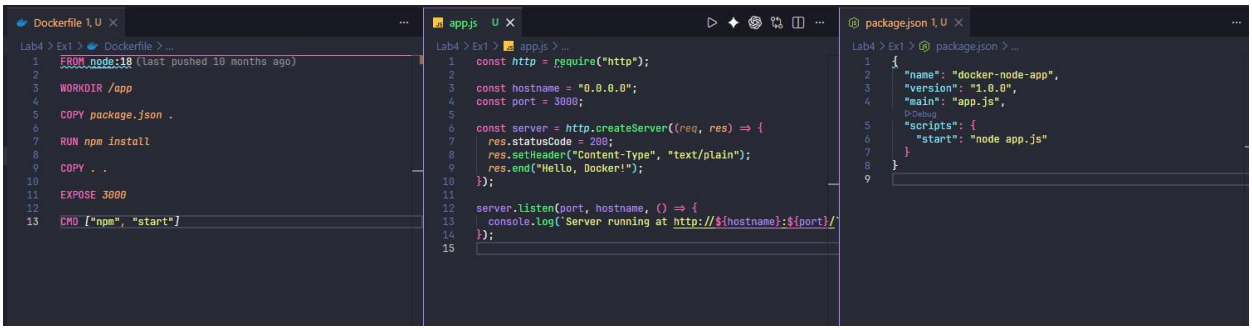
31 docker run -d -p 8080:80 my_nginx_image

```
D:\HK2_2026\KTHT\KTTKPM_HuynhThanhPhuong_22671361_DHKTPM18ATT\Lab4\Ex30 [main ≡ +2 ~0 -0 !]> docker run -d -p 8080:80 my_nginx_image
6b26e33e5dad2b852c653d5fb5a064e52089a4ced83b9429cfa0a5251636d042
D:\HK2_2026\KTHT\KTTKPM_HuynhThanhPhuong_22671361_DHKTPM18ATT\Lab4\Ex30 [main ≡ +2 ~0 -0 !]> docker ps
CONTAINER ID   IMAGE            COMMAND                  CREATED         STATUS         PORTS                                       NAMES
6b26e33e5dad   my_nginx_image   "/docker-entrypoint.…"   3 seconds ago   Up 3 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   hopeful_hawking
```

Phần 2: Thao tác với Dockerfile

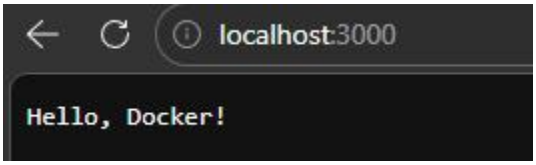
Bài 1: Tạo Dockerfile chạy một ứng dụng Node.js đơn giản Yêu cầu:

Viết Dockerfile để chạy một ứng dụng Node.js hiển thị "Hello, Docker!" trên cổng 3000. Sử dụng node:18 làm base image.




```
D:\HK2_2026\KTHT\KTTKPM_HuynhThanhPhuong_22671361_DHKTPM18ATT\Lab4\Ex1 [main ≡ +3 ~1 -0 !]> docker build -t node_hello .
[+] Building 12.8s (4/9)                                                                          docker:desktop-linux
 => [internal] load metadata for docker.io/library/node:18                                                         1.0s
 => [internal] load .dockerignore                                                                                  0.0s
 => => transferring context: 2B                                                                                    0.0s
 => [1/5] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783   11.7s
```

```
D:\HK2_2026\KTHT\KTTKPM_HuynhThanhPhuong_22671361_DHKTPM18ATT\Lab4\Ex1 [main ≡ +3 ~1 -0 !]> docker run -p 3000:3000 node_hello

> docker-node-app@1.0.0 start
> node app.js

Server running at http://0.0.0.0:3000/
```



Bài 2: Tạo Dockerfile chạy một ứng dụng Python Flask

Yêu cầu:

Viết Dockerfile để chạy một ứng dụng Flask hiển thị "Hello, Docker Flask!" trên cổng 5000.

Sử dụng python:3.9 làm base image.

```
D:\HK2_2026\KTHT\KTTKPM_HuynhThanhPhuong_22671361_DHKTPM18ATT\Lab4\Ex2 [main ≡ +4 ~1 -0 !]> docker build -t flask_app .
[+] Building 1.5s (4/9)
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 206B
 => [internal] load metadata for docker.io/library/python:3.9
 => [internal] load .dockerignore
```

Dockerfile 1, U

Lab4 > Ex2 > Dockerfile > ...

```
FROM python:3.9 (last pushed 4 months ago)

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 5000

CMD ["python", "app.py"]
```

app.py U

Lab4 > Ex2 > app.py > ...

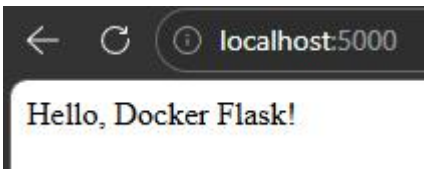
```
from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    return "Hello, Docker Flask!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

app.py U | requirements.txt U

Lab4 > Ex2 > requirements.txt

```
flask
```

```
D:\HK2_2026\KTHT\KTTKPM_HuynhThanhPhuong_22671361_DHKTPM18ATT\Lab4\Ex2 [main ≡ +4 ~1 -0 !]> docker run -p 5000:5000 flask_app
 * Serving Flask app 'app'
 * Debug mode: off
```



Bài 3: Tạo Dockerfile chạy một ứng dụng React

Yêu cầu:

Viết Dockerfile để build và chạy một ứng dụng React.

Sử dụng node:18-alpine làm base image.

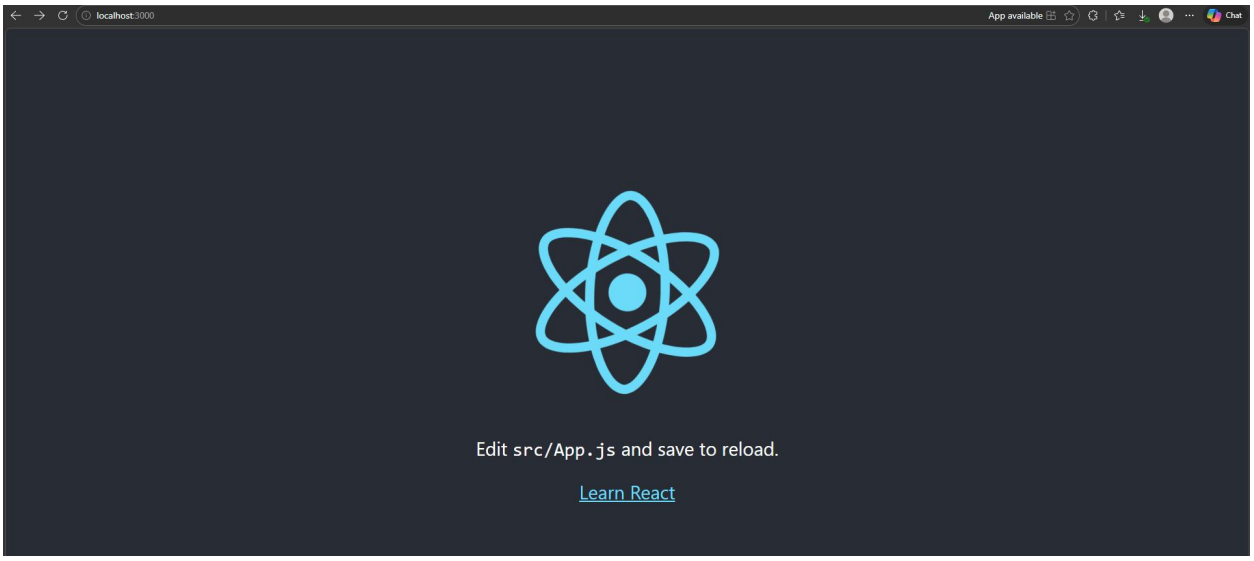
Dockerfile 2, U

Lab4 > Ex3 > my-app > Dockerfile > ...

```
# Stage 1: Build React app
FROM node:18-alpine AS build (last pushed 1 year ago)

WORKDIR /app

COPY package*.json ./
RUN npm install

COPY . .
RUN npm run build


# Stage 2: Serve with Nginx
FROM nginx:alpine (last pushed 2 days ago)

COPY --from=build /app/build /usr/share/nginx/html

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

```
D:\HK2_2026\KTHT\KTTKPM_HuynhThanhPhuong_22671361_DHKTPM18ATT\Lab4\Ex3\my-app [main ≡ +5 ~1 -0 !]> docker build -t react_app .
[+] Building 7.0s (5/13)                                                                                    docker:desktop-linux
 => [build 1/6] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e   5.9s
 => => extracting sha256:dd71dde834b5c203d162902e6b8994cb2309ae049a0eabc4efea161b2b5a3d0e                                    0.6s
 => => sha256:3370263bc02adcf5c4f51831d2bf1d54dbf9a6a80b0bf32c5c9b9400630eaa08 20.25MB / 20.25MB                             2.9s
```

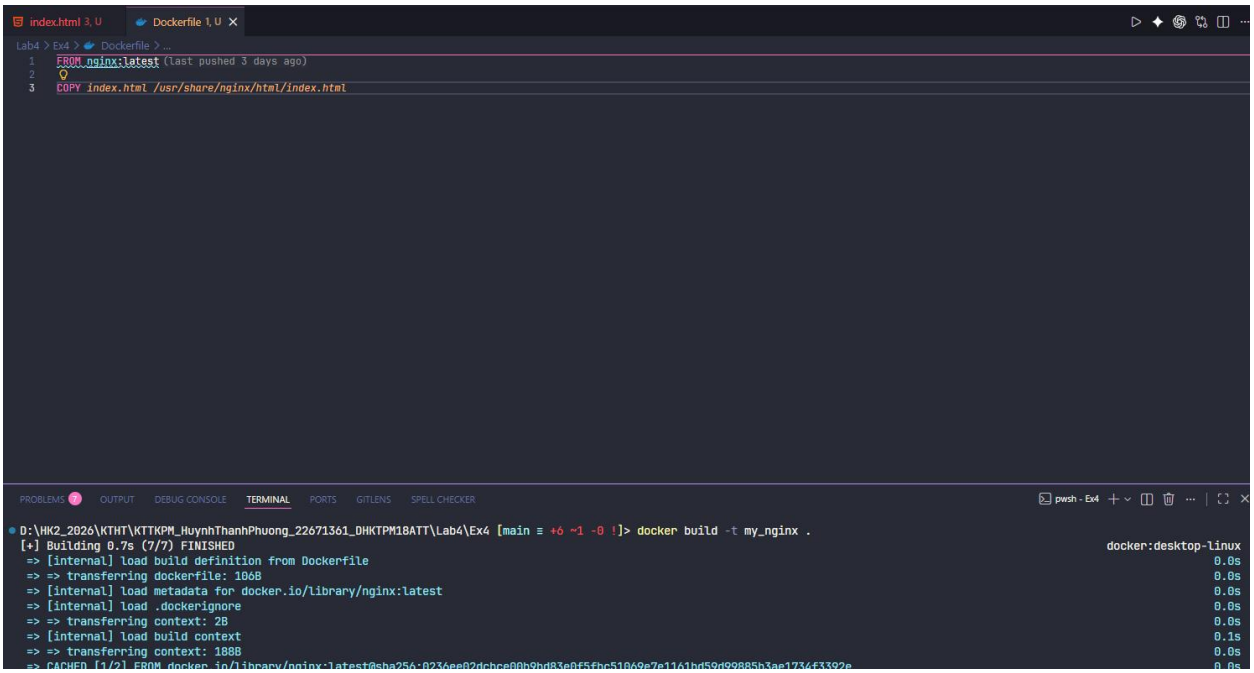
```
D:\HK2_2026\KTHT\KTTKPM_HuynhThanhPhuong_22671361_DHKTPM18ATT\Lab4\Ex3\my-app [main ≡ +5 ~1 -0 !]> docker run -p 3000:80 react_app
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
```


localhost:3000
App available
Chat
Edit src/App.js and save to reload.
Learn React

Bài 4: Tạo Dockerfile chạy một trang web tĩnh bằng Nginx

Yêu cầu:

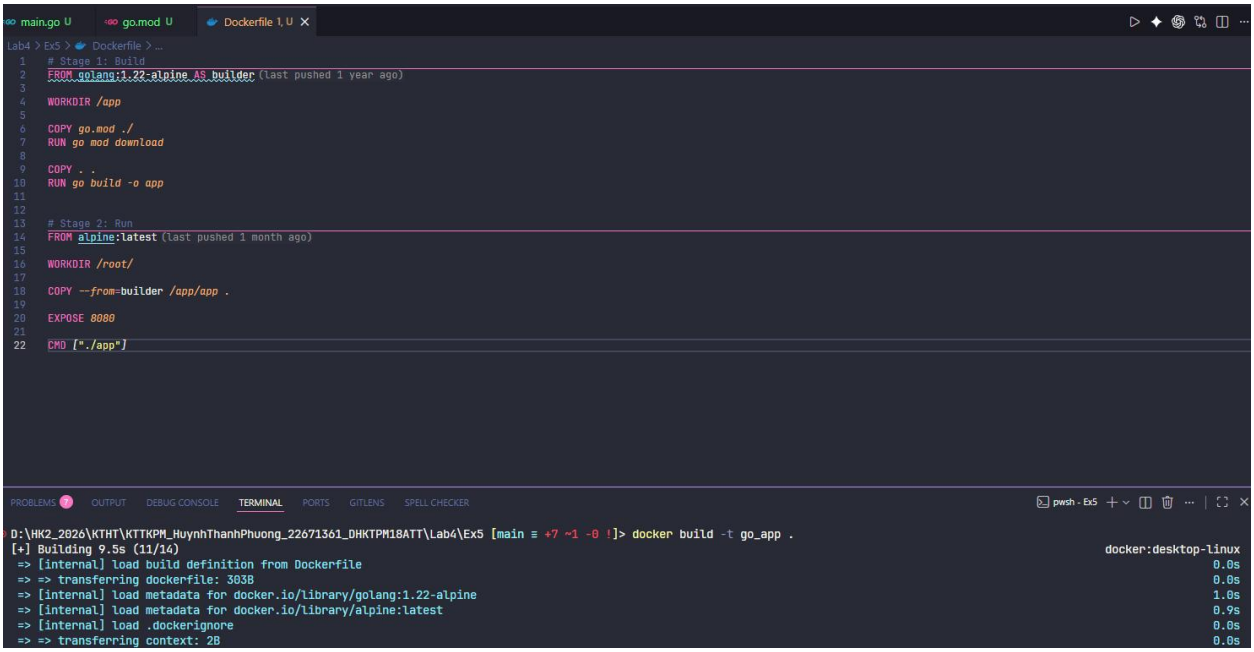
Tạo một file index.html đơn giản và sử dụng nginx:latest để phục vụ trang web.

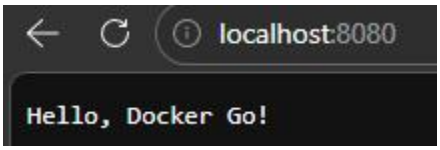


Bài 5: Tạo Dockerfile cho ứng dụng Go

Yêu cầu:

Viết Dockerfile để build và chạy một ứng dụng Go đơn giản.



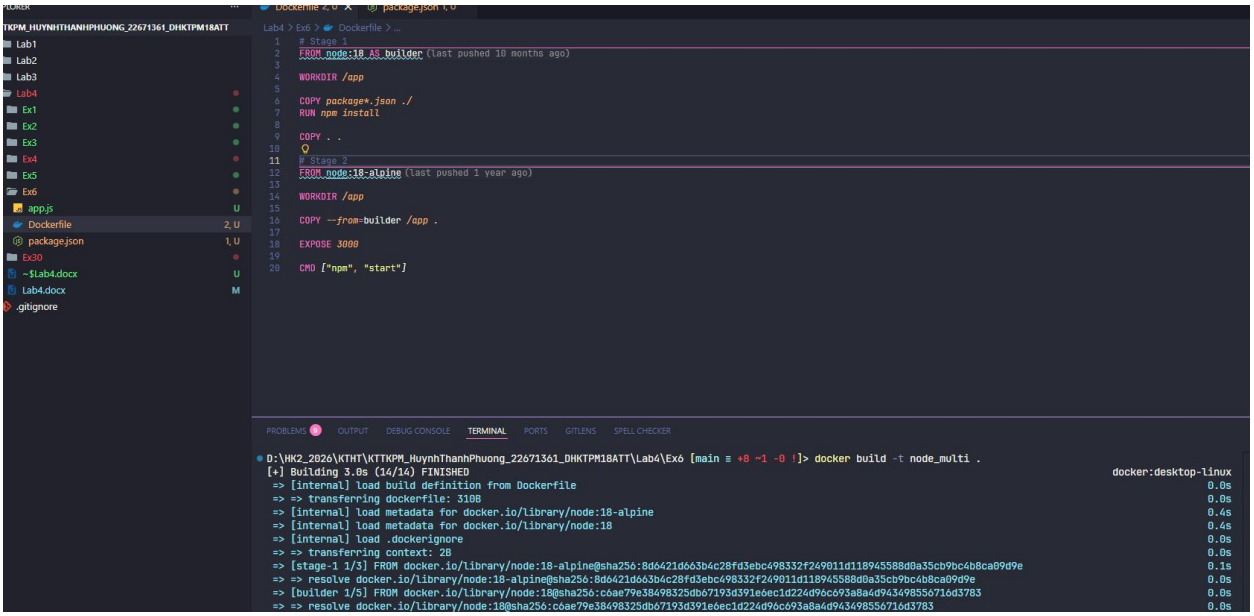


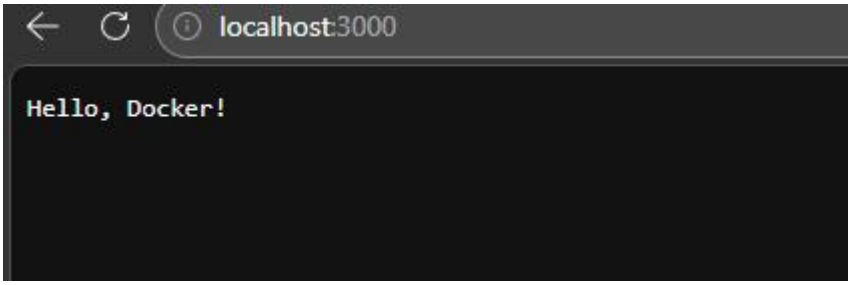
Bài 6: Sử dụng Multi-stage Build trong Dockerfile

Viết Dockerfile để build một ứng dụng Node.js với hai stage:

Stage 1: Dùng node:18 để build code.

Stage 2: Dùng node:18-alpine để chạy ứng dụng đã build.



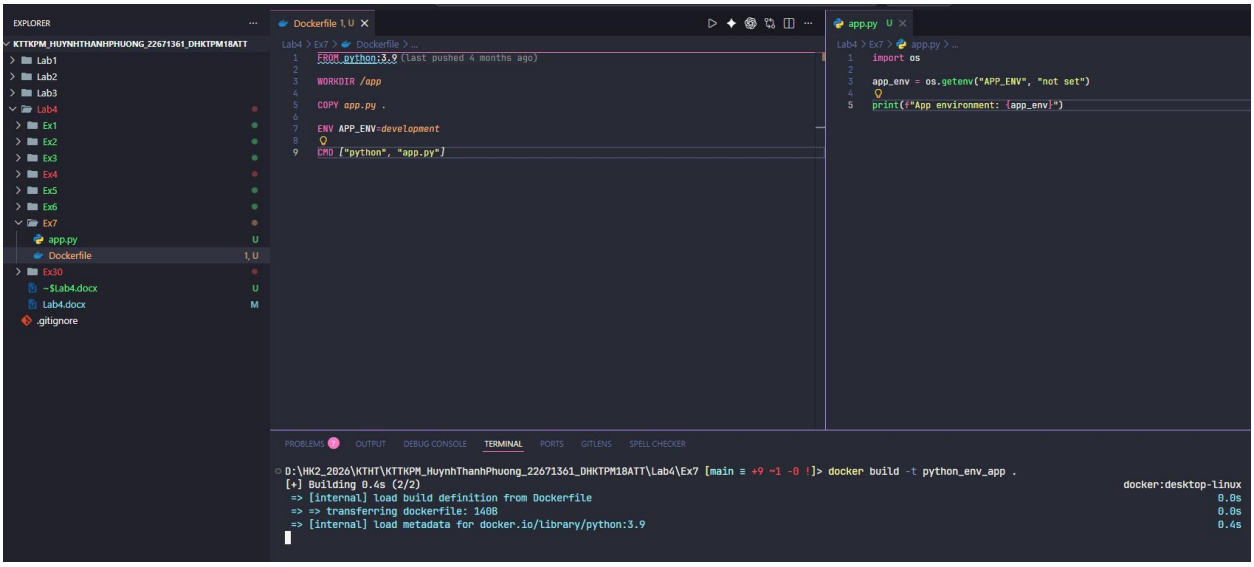


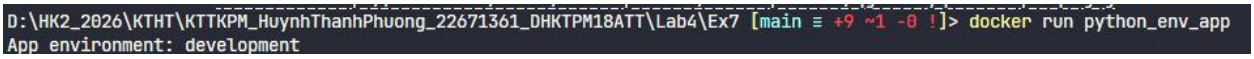
Bài 7: Sử dụng biến môi trường trong Dockerfile

Yêu cầu:

Viết Dockerfile cho ứng dụng Python đọc biến môi trường APP_ENV và in ra màn hình.

Sử dụng ENV APP_ENV=development trong Dockerfile.



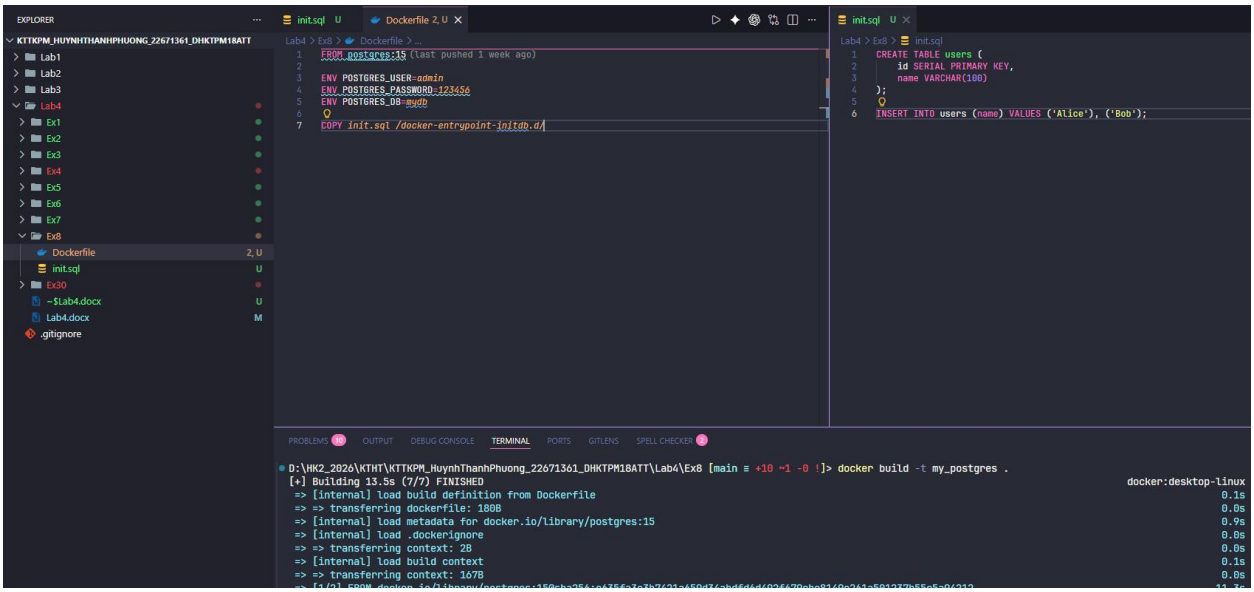


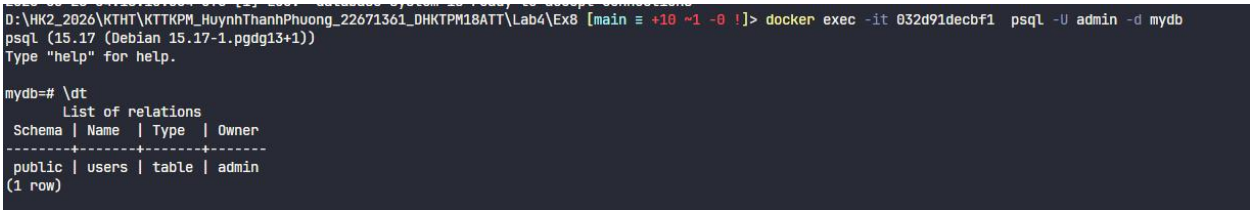
Bài 8: Tạo Dockerfile cho PostgreSQL tùy chỉnh

Yêu cầu:

Viết Dockerfile để chạy PostgreSQL (postgres:15).

Thêm file SQL để tự động tạo database khi container chạy lần đầu tiên.



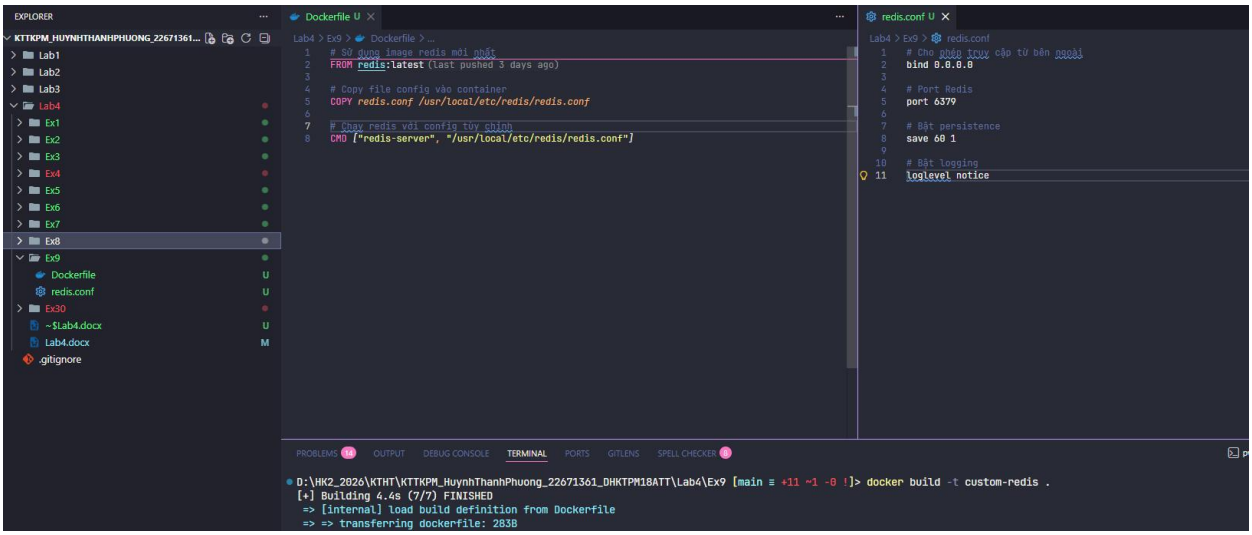


Bài 9: Tạo Dockerfile chạy Redis với cấu hình tùy chỉnh

Yêu cầu:

Viết Dockerfile sử dụng redis:latest.

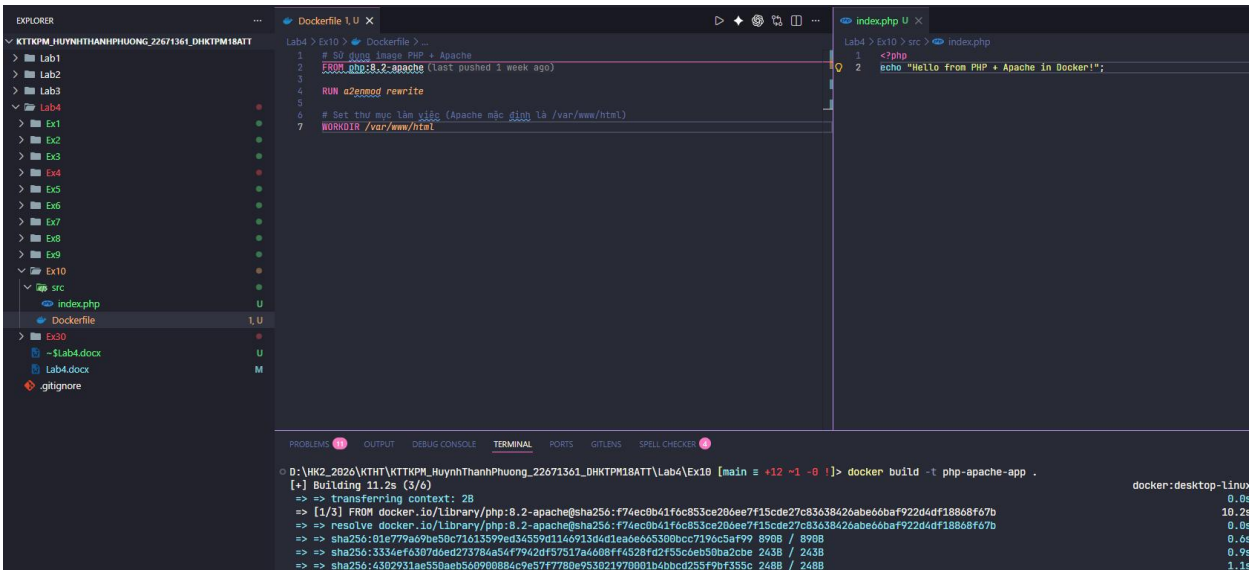
Thêm file redis.conf vào container.




```
D:\HK2_2026\KTHT\KTTKPM_HuynhThanhPhuong_22671361_DHKTPM18ATT\Lab4\Ex9 [main ≡ +11 ~1 -0 !]> docker exec -it my-redis redis-cli
127.0.0.1:6379> ping
PONG
```

Bài 10: Chạy ứng dụng PHP với Apache

Yêu cầu:

Viết Dockerfile để chạy một ứng dụng PHP đơn giản (php:8.2-apache).

Mount mã nguồn từ máy host vào container.